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Multi Mini-Frac for In-Situ Stress Evaluation Using Real-Time Intelligent Coiled Tubing and Inflatable Straddle Packer a Field Case Study

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Abstract

In subsurface environments with limited data availability, accurate in-situ stress characterization is crucial for optimizing future reservoir development, particularly in appraisal wells where formation properties remain unknown. This case study presents a methodology for conducting multi mini-frac tests to determine the horizontal stress of the formation in a newly drilled appraisal well.

The approach leverages a custom-engineered inflatable straddle assembly, deployed via Intelligent Coiled Tubing Technology (ICTT) to ensure precise depth placement and enhanced operational control, both critical for the safe and efficient execution of the procedure.

A 140-meter-long tubing-conveyed perforation (TCP) was executed using ICTT, enabling controlled perforation within the target interval and real-time operational monitoring. Given the necessity of conducting multiple mini-frac stages across such an extensive perforated section, a four-stage custom-designed inflatable straddle assembly was implemented, incorporating:

- Inflation/Deflation Control,
- Displacement or Circulation,
- Pressure Shut-in (Formation/Inflation Monitoring),
- Test Zone Exposure to the Work String for real-time pressure bleed-off assessment.

The straddle assembly, equipped with reinforced rubber elements, was set in the perforated interval, an area inherently prone to mechanical deformation and damage to rubber elements due to liner conditions post-perforation.

To capture critical formation response parameters, a dual-gauge carrier was integrated between the packer elements, embedding memory gauges with a 96-hour operational lifespan to sustain extended test durations. Additionally, check valves were removed from the ICTT motorhead assembly, enabling direct formation response measurement and precise control of post-frac leak-off dynamics. A low-viscosity, non-crosslinked product was selected to accurately determine the true horizontal stress of the formation, ensuring fracture

sustainability without premature closure. The absence of proppant or solid particulates was mandatory due to the intrinsic operational constraints of the straddle packer, which cannot accommodate solid-laden fluids.

However, the use of a low-viscosity fluid system also introduced operational risks, particularly with fluids utilised in the process of inflating the packers. This necessitated the integration of a contingent deflation mechanism, and precise fluid control strategies to mitigate risks and ensure the reliable operability of the inflatable straddle assembly

The multi mini-frac tests provided clear formation data, enabling a determination of in-situ horizontal stress. The use of real-time ICTT facilitated continuous monitoring and precise operational control, enhancing data acquisition reliability throughout the process.

Despite the operational constraints of using a solid-free, the low-viscosity fluid successfully initiated and sustained the fracture while gradually dissolving over time, preventing premature closure. The additional deflation mechanism remained a contingency measure, ensuring operational safety and packer retrieval efficiency in case of unexpected situation.

Introduction

The accurate characterization of the in-situ stress field is a fundamental step in the planning and execution of geothermal wells drilled into sedimentary formations at intermediate depths, typically between 1 500 and 3 000 meters. Target formations of Permian age—such as the Rotliegend Group, and specifically the Slochteren Sandstone—exhibit moderate to high porosity with permeability ranging from a few millidarcy up to several hundred millidarcy, depending on diagenetic cementation, depositional facies, and burial history. While not strictly low-permeability, many of these intervals still require hydraulic stimulation to achieve sufficient flow rates for geothermal production.

In this context, knowledge of the horizontal stress state—and specifically the minimum horizontal stress (Sh_{min})—is essential for assessing stimulation feasibility, predicting fracture geometry, and ensuring wellbore stability. Accurate estimation of Sh_{min} is also critical to evaluate fracture containment and vertical propagation risk, particularly in stratified lithologies with contrasting mechanical properties. In this basin, Sh_{min} is typically assumed to correspond to the fracture gradient (FG), with values derived from offset LOT and FIT data. However, such estimates are inherently uncertain due to lateral lithological variability and completion-specific factors, making direct measurement through field testing highly desirable.

Pore pressure (PP) in the region is generally hydrostatic, with no overpressure expected. Pressure gradients vary with depth and lithostratigraphy: approximately 1.0 SG equivalent mud weight (EMW) in shallow freshwater aquifers, rising to 1.05–1.08 SG EMW in deeper units including the Zechstein, Rotliegend, and Limburg Groups. For the Slochteren Formation, hydrostatic conditions were assumed based on regional analogues and offset well data.

Mini-fracture tests in extended leak-off mode (XLOT) are among the most reliable field techniques for constraining in-situ stress. Unlike simple leak-off tests, XLOTs involve multiple cycles of fracture closure and reopening, providing robust estimates of breakdown pressure, reopening pressure, and closure pressure. These pressures directly correspond to Sh_{min} and, when integrated with geological and geophysical data, yield insights into stress anisotropy and mechanical behavior.

In the present case, the target interval corresponds to the Slochteren Formation, characterized by fine to medium-grained sandstones interbedded with claystone and minor evaporitic or carbonate-rich layers. Porosity typically ranges from 10% to 25%, with permeability spanning sub-millidarcy to over 1 Darcy depending on local facies and cementation. The formation's brittle nature, moderate-to-high elastic moduli, and lateral heterogeneity limit the reliability of indirect stress estimation methods (e.g., poroelastic models or log-derived correlations). Hence, a direct minifrac-XLOT approach was adopted to constrain Sh_{min} and calibrate the geomechanically model under site-specific subsurface conditions.

Technically robust XLOT execution requires control of injection rates, monitoring of pressure transients, and analysis of key parameters such as fracture closure pressure (P_{cl}), fluid efficiency, and pressure decline during shut-in. These are highly sensitive to operational setup, formation properties, and well geometry. Given borehole constraints in this well, an inflatable packer was used to isolate the test interval—details of this setup will be presented in the dedicated Methods section.

This campaign is part of a larger national geothermal exploration effort aimed at reducing subsurface uncertainty in underexplored areas. The program focuses on acquiring high-resolution geological, geophysical, and geomechanically information to support sustainable energy initiatives. Within this strategic framework, the XLOT operation reported here serves as a targeted stress characterization campaign, designed to assess stimulation feasibility and strengthen well planning strategies in frontier geothermal regions.

Operational Framework and Testing Strategy

The post-drilling campaign was developed with the dual goal of technical rigor and operational efficiency, leveraging a fully rig-less intervention strategy. Once the drilling and completion phases were finalized, the rig was demobilized and the subsequent testing program—including a multi-zone minifrac/XLOT operation—was executed entirely via coiled tubing. This strategy allowed the project to significantly reduce operational time, minimize surface impact, and maintain economic viability in a geothermal context where well interventions must often balance high technical standards with strict budget constraints.

From a geomechanical standpoint, the primary objective of the campaign was the in-situ quantification of the minimum horizontal stress (Sh_{min}) across multiple depths within the target reservoir unit. This was not only aimed at supporting future stimulation design, but also at validating the local stress model and assessing intra-reservoir stress variability—a critical factor in heterogeneous formations such as the Permian sandstones involved in this case.

To meet these goals under rig-less conditions, the testing strategy was structured around the execution of repeated, controlled XLOTs across at least four distinct perforated intervals, all within a single coiled tubing deployment. This required the integration of an intelligent CT system, equipped with a real-time sensing bottom-hole assembly, capable of acquiring downhole pressure data and supporting precise tool positioning and monitoring the setting sequence. Surface and subsurface telemetry were used in tandem to ensure operational control and accurate synchronization throughout each test stage.

At the core of the system were both the intelligent coiled tubing technology and the CT-conveyed inflatable straddle packer system, jointly engineered to enable selective isolation of each cased-hole test interval during a single run-in-hole. Operating in a perforated casing environment added significant complexity: the presence of exposed perforation tunnels posed a tangible risk of puncture to the rubber elements, potentially compromising the packer's ability to maintain pressure integrity or execute multiple inflation-deflation cycles. In the worst-case scenario, it could lead to partial extrusion of the rubber into the perforation cavity, causing the inflatable packer to get stuck in hole even after the deflation process.

The system was capable of:

- Selective interval isolation with mechanical integrity despite potential casing damage from perforations;
- Precise fluid injection and shut-in control, free from external interference;
- Compatibility with viscous fluids designed to support controlled fracture propagation without compromising tool function;
- Real-time downhole pressure monitoring, enabling high-resolution tracking of breakdown, reopening, and closure events;

- Repeatability of the test cycle, ensuring multiple stress measurements could be acquired without the need for re-mobilization.

Each test sequence was executed following a pre-validated protocol:

1. Low-rate injection to characterize wellbore storage and near-well response;
2. Step-rate analysis to detect fracture initiation;
3. Multiple injection–shut-in cycles to observe fracture reopening and closure behavior under stable conditions;
4. Post-test flowback and bleed-down to ensure safe deflation and recovery of the inflatable packer.

Due to the presence of narrow internal flow paths and inflation channels in the straddle packer design, fluid selection played a critical role in the overall operability of the system. The fracturing fluid—designed for minimal filtration residue and low-to-moderate viscosity—was selected to balance fracture effectiveness with tool compatibility, as detailed in the following chapter.

This integrated approach enabled the acquisition of high-quality stress data across four distinct zones within a single coiled tubing deployment. The result is a fully rig-less, multi-layer XLOT methodology, specifically tailored for geothermal applications, where maximizing information density per intervention is paramount. It represents a step change in how stress testing can be conducted in challenging operational environments—combining pressure fidelity, zonal resolution, and cost-efficiency in a unified workflow.

IPI Inflatable Straddle Packer Design

For this application, the HP ST114 multipurpose tool was deployed. It offers a versatile and efficient solution for a wide range of downhole operations, including fracturing, and is ran on a straddle packer system which is adaptable to different borehole geometries and borehole conditions. The HP ST114 tool family includes multi-cycle inflatable packer systems featuring a four-stage operating mechanism: packer inflation/deflation, annular circulation, shut-in isolation, and interval treatment.

In this deployment, the tool was configured with a dual-packer straddle assembly, equipped with custom-made dual gauge carriers. The HP Straddle System was fitted with 5" contra-wound High Pressure Packer Elements, each reinforced with four internal layers of high strength steel wires to withstand injection pressures up to 10,000 psi. The tubeless heavy-duty rubber cover was specifically designed to mitigate jamming during setting within perforated sections, while maintaining seal integrity even under high-pressure frac conditions.



Figure 1—Cross Section of the 5" element design and construction.

The packer size and configuration were selected through detailed application design, mathematical modeling and simulations to ensure both pressure withstanding and post-frac retrievability. This prevented packer rubber from extruding into perforation tunnels, which could otherwise cause jamming or tool sticking. The internal inner rubber layer was engineered with sufficient wall thickness to allow multiple inflation-deflation cycles, enabling the execution of up to eight zone stimulations in a single CT run.

The internal filtration system was also modified for this operation. Larger-aperture screens were incorporated to accommodate viscous gel-based fluids, while a repositioned intake pipe created a sump effect in the upper section, trapping oversized particles and minimizing the likelihood of tool clogging.

During the circulation stage, coiled tubing fluids could be reversed to the annulus, conditioning the coil by displacing undesired fluids. This ensured that only clean, dedicated frac fluid was injected into the formation, improving stimulation efficiency and data quality.

The tool's balanced piston design provided a zero-displacement setting mechanism, eliminating tool movement due to pressure shifts. Additionally, the tool system features an inflation bypass system that vents the test zone between straddle packers during packer inflation thus preventing displacement of well fluid into the formation or potential "squeezing" due to packer inflation. These two features allow for more accurate and representable data, as minimal pressure spikes or pressure drops occur as a result of operating the tool.

A key feature for data quality assurance was the shut-in valve installed above the upper packer, which can be closed after the mini frac, therefore stopping pumping through the CT, isolating the reservoir from hydrostatic fluid invasion. Without this feature, post-frac leak-off data would have been distorted by continued influx from the wellbore column, jeopardizing the capture of an accurate pressure dropdown curve. The tool string was also equipped with two memory gauges—one primary and one backup—mounted on extension carriers between the packer elements. This arrangement enabled precise monitoring of fracture closure pressure during the fall-off phase, decoupled from tool-body dynamics and shielded from mechanical interference.

After each fracture stage, the ST tool was actuated to the deflation stage, discharging the inflation fluid into the well annulus. The contra-wound, wire-reinforced element allowed full-diameter retraction without any plastic deformation on the packer element ends, which is a typical inherent flaw of the slat-type inflatable packers design available in the market. This best in class diameter recoverability ensures smooth, reliable retrieval and allowing subsequent zones to be tested in sequence down up in the same trip.



Figure 2—HP ST114 with 5in Straddle System

Fracturing Fluid Selection and Rheological Performance

The success of an XLOT operation depends not only on the mechanical design and pressure monitoring system, but also critically on the selection of a suitable fracturing fluid. In the specific operational context discussed here, the fluid needed to fulfill multiple constraints simultaneously: it had to ensure adequate leak-off control in the formation, possess sufficient viscosity for effective fracture initiation and propagation, and, remain pumpable through the restricted internal flow paths, orifice selected was a 5 mm, of the inflatable straddle packer system, in particular when passing through the stages mechanism flow path very small.

The fluid selected for the test was a modified polymeric gel "L" version, a lower-viscosity and custom-made version of the base product, as the higher viscosity "X" version failed to pass the 5-micron filtration

test and could not be reliably pumped through the small internal channels of the straddle packer mechanism, potentially compromising inflation and deflation cycles. The "L" version, on the other hand, retained favorable rheological properties while ensuring full compatibility with the packer's narrow ports and met all filtration criteria required for safe deployment in a constrained tool geometry.

The product was used at a concentration of 36% premixed solution, as determined in pre-job lab testing. This concentration provided a controlled rheological profile suitable for pressure transmission, while maintaining non-thixotropic behavior—a key requirement to avoid unintentional fluid retention within the rubber inflation elements of the packer. Excess gel retention in these areas could have compromised the pressure bleed-off phase and packer retrieval, representing a non-negligible operational risk.

From a rheological standpoint, the "L" gel version exhibits pronounced shear-thinning behavior, as illustrated in Figure 2, where it is compared against a previously used drilling mud—employed within this scope of work—that was originally formulated to enhance filter cake deposition during surface injection operations. While the "classic fluid Mud" exhibits higher apparent viscosity across all shear rates, its elevated rheological resistance and filtrate characteristics made it unsuitable for tool internal geometry constrained. In contrast, "L" version Gel achieves a balance between viscosity and pumpability, particularly at low-to-moderate shear rates ($1\text{--}100\text{ s}^{-1}$), which are most representative of flow conditions and across the inflatable packer assembly.

The use of "L" version gel enabled the team to:

- Maintain pressure stability throughout the injection and shut-in phases;
- Prevent early leak-off and optimize pressure signature clarity;
- Avoid obstruction in the straddle packer's inflation system;

Physicochemical data confirm a stable fluid with a density of approximately 1.15 g/cc , pH range between 5–8, and high thermal stability. Importantly, the product is non-reactive under downhole conditions and exhibits minimal environmental impact (classified Aquatic Chronic 3)

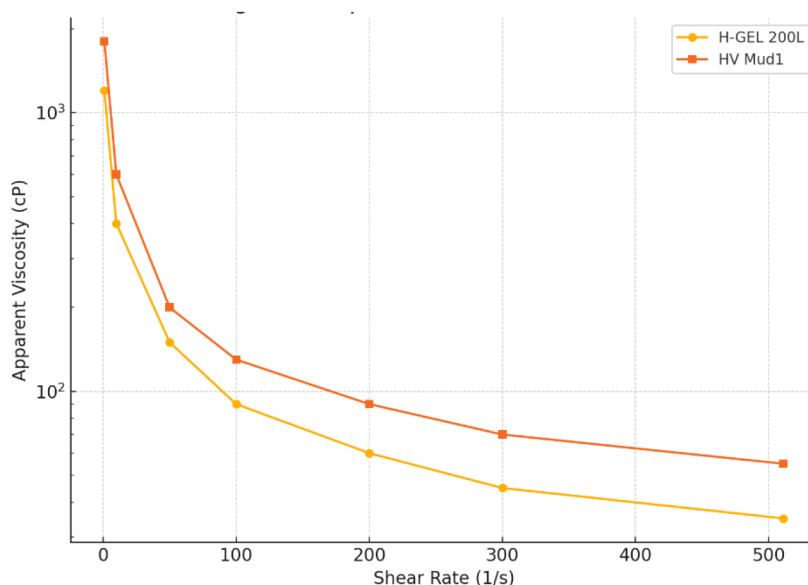


Figure 3—Rheological Comparison "L" version Gel vs Drilling Fluid Mud previously used.

Intelligent Coiled Tubing Technology

The deployment of the straddle packer assembly was performed using an Intelligent Coiled Tubing (CT) system, specifically engineered to combine real-time downhole telemetry with high-pressure intervention capabilities in a fully rig-less operational setup. This configuration was selected to meet the demands of the XLOT campaign, enabling multi-zone fracture execution with precise depth control, and allowing continuous monitoring of critical downhole parameters during all operational phases—including run-in-hole, packer setting, injection, shut-in, and deflation.

At the core of the system was a 4 mm hybrid coaxial cable, combining a 17 AWG copper conductor for power and data transmission via power-line communication, with a Fiber in Metal Tube (FIMT) containing two single-mode and two multi-mode optical fibers. Although the fiber channels were not utilized for distributed sensing in this specific application, their integration ensured robustness, bandwidth capacity, and compatibility with future high-resolution geothermal diagnostics under high-temperature conditions.

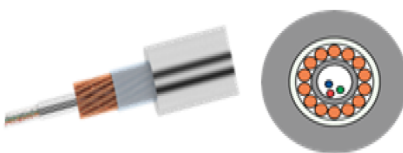


Figure 4—Intelligent Coiled Tubing Cable Configuration.

The sensing bottom-hole assembly (BHA) provided a comprehensive package of high-accuracy instrumentation. It included:

- Internal and external pressure sensors, with a resolution of $\pm 0.02\%$ F.S. and pressure ratings up to 15,000 psi, ensuring high-fidelity pressure acquisition during inflation and frac phases;
- Temperature sensors with 1 °C resolution and a range of -40 to $+177$ °C, enabling real-time thermal tracking during both fluid displacement and static phases;
- An Active Casing Collar Locator (ACCL) capable of precise collar detection even at low CT velocities and in large-diameter completions, where conventional magnetic tools may fail, providing high-confidence depth correlation and enabling accurate positioning of the straddle assembly in perforated intervals;
- A tension and compression module with 20 lbs resolution and 50,000 lbs range, used to monitor weight-on-bit (WOB) and identify mechanical responses during inflation. This allowed operators to detect strain variations correlated to rubber expansion and set confirmation;
- Inclination, vibration, and torque sensors completing the tool's diagnostic capabilities and providing a full picture of toolstring dynamics throughout the run.

The real-time telemetry enabled direct observation of delta pressure across the straddle packer rubber elements during run-in-hole and inflation, validating the packer set sequence. Furthermore, real-time monitoring allowed immediate detection of any irregularities during pumping or fracture propagation, supporting on-the-fly decision-making process.

An essential component of the intelligent coiled tubing technology is the surface acquisition and visualization software, which plays a critical role in enabling real-time decision-making during downhole operations. In this application, the system was used to interface with the sensing BHA, allowing continuous data acquisition and display of key downhole parameters.

One particularly valuable functionality was the log-up/log-down signal processing feature, which allowed the operator to perform high-precision depth correlation by comparing casing collar locator responses in both upward and downward tool movement. This enhanced signal clarity, helped eliminate transient noise,

and enabled the clear identification of collar spikes, even in large-diameter completions where signal-to-noise ratio is often degraded.

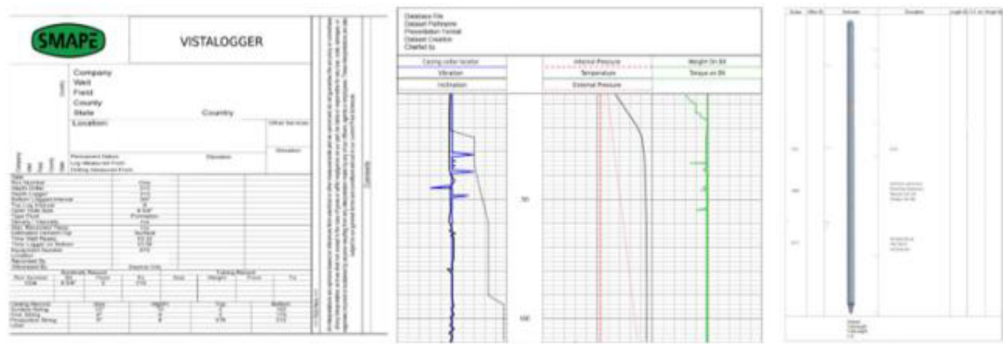


Figure 4—Intelligent Coiled Tubing Software.

Execution

The mini-frac operations were executed as planned, with the straddle packer system successfully inflated and deflated five times within a single coiled tubing run, without encountering any operational issues. This result is particularly noteworthy given the pre-existing concerns related to casing perforations, which in previous operations had led to premature rubber element failure and CT entrapment.

The packer assembly was configured with a 5 mm bleed orifice, slightly larger than originally specified, in order to improve and expedite the deflation performance in wells with low annulus fluid levels. The enlarged orifice was specifically designed to facilitate rapid pressure equalization across the inflatable elements, allowing for more reliable rubber retraction during packer deflation stage. Based on hydraulic modeling, the anticipated pressure drop across the orifice was approximately 1,000 psi at a flow rate of 0.45 bpm, and 1,500 psi at 0.86 bpm. In addition, the tool was configured with an emergency deflate system, which provided contingency in case of any difficulties in deflating the straddle packers.

The tools was configured with the shear pin value shown in table 1, taking into account also tolerance value of +/- 15% for the shear rating.

Table 1—Emergency Deflation Setting.

Description	Activation method	Number of pins	Rating Per pin	Total Rating
Emergency Deflation	Pressure/ Pull	6	1,848 psi 4,780 lbs	12,936 psi 28,680 lbs

Due to prior full wellbore accessibility confirmed in earlier runs, the tool was deployed in a closed-loop circulation mode. Specific contingencies were considered during run-in-hole, particularly to address any unexpected slack-off conditions that could result from pressure imbalances or premature element expansion. In such events, corrective actions included halting CT movement, applying controlled overpulls, and monitoring fluid behavior through the weep hole to allow gradual deflation and packer recovery. During the operation, the tool didn't experience any challenges while running in hole.

Upon approaching the first target depth, real-time depth correlation was performed using the sensing BHA's high-resolution ACCL. The process was enhanced by the log-up and log-down capabilities of the Warrior acquisition software, which allowed for effective noise filtering and spike signal enhancement —particularly valuable in large-diameter completions where collar signal clarity is often degraded. This approach enabled accurate identification of casing collars and reliable confirmation of tool positioning relative to the designated perforated interval. Once the straddle assembly was aligned with the target zone, the coiled tubing string was set to a neutral weight condition, and the packer inflation sequence was initiated.

Inflation was initiated by pumping at a controlled flow rate through the coiled tubing, starting at approximately 0.7 bpm, gradually increasing pressure to around 1,000 psi. The rate was slowly increased to 0.86 bpm, allowing pressure to build toward the 1,500 psi target. The packer inflation and setting was confirmed by a change in coiled tubing weight and pressure readings, observed both at surface and more precisely through the real-time downhole sensing BHA, which continuously tracked tension and compression forces, as well as internal and external pressures. The variation in axial load was directly correlated to the inflation of the elements, providing a confirmation of straddle packer setting.

Once target inflation pressure was reached, the coiled tubing was slacked off to maintain a weight-on-bit between 4,000 and 5,000 lbs, a critical parameter to confirm tool anchoring and effective mechanical sealing against the casing wall. The applied compressive load was constantly monitored in real time via the tension/compression module of the sensing BHA, allowing the surface crew to verify packer engagement and stability before proceeding to the injection phase.

Following inflation, the tool was mechanically stroked into the next operational stage (see Table 2) by advancing the internal mandrel. During this transition, pressure bleed-off was monitored as a diagnostic indicator of proper stage alignment, as the tool strokes past the circulation stage. Importantly, pumping was never interrupted during this phase, in order to avoid unintended packer deflation due to loss of internal pressure. Instead, controlled CT movement—monitored in real time via the downhole sensing BHA—enabled the internal shifting of the straddle's flow path, thereby redirecting the fluid path from the inflatable packers to the treatment interval between the inflatable packers.

Once the tool reached the correct position and the flow path opened between the packer elements, the mini-frac sequence commenced. Only at this confirmed configuration was the viscous fluid introduced through the coiled tubing. This procedural step was critical to avoid pressurizing the inflation system with viscous fluid, which could have jeopardized the packer deflation process due to fluid retention or filter plugging.

Fracturing fluid was pumped at constant rate, ensuring controlled fracture propagation and stable pressure development. Each mini-frac stage was limited to a maximum of 12 m³ of viscous gel, after which the system was flushed with water. A precisely calculated water displacement volume, corresponding to 1.1 times the internal CT capacity, was used to guarantee complete displacement of the gel before transitioning to the next stage. This ensured that no residual viscous fluid remained in the internal flow paths of the ST tool during subsequent deflation and repositioning phases, thus preserving tool integrity and repeatability across multiple test intervals.

Table 2—Straddle Packer Stages.

STAGES	DESCRIPTION	STROKE LENGTH mm (inches)
1 st Stage (Datum)	Packer Inflation/Deflation	0mm
2 nd Stage	Displacement or circulating above the straddle system, while shut in	254mm (10")
3 rd Stage	Pressure shut in (formation/inflation)	254mm (10")
4 th Stage	Test zone is open to the work string	254mm (10")

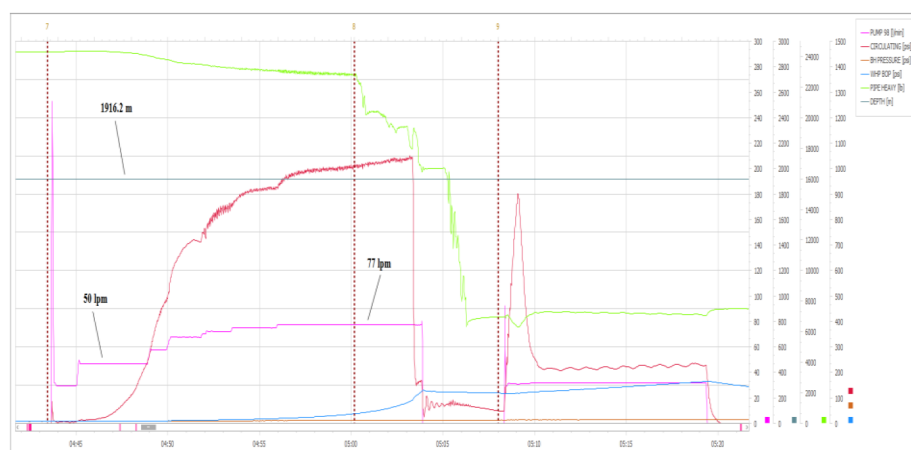
Upon completion of each mini frac, the ST tool was stroked to the pressure shut in stage to isolate the CT from the test interval and allow pressure build-up monitoring. Throughout this phase, the memory gauges installed between the packer elements recorded the fall-off profile, capturing pressure decline in the fracture until closure could be identified.

At the end of each stage, the mandrel was pulled back to the deflation position, initiating passive pressure bleed-down through the weep hole. A minimum wait time of 20 minutes was required to ensure full retraction of the rubber elements before continuing to the next zone.

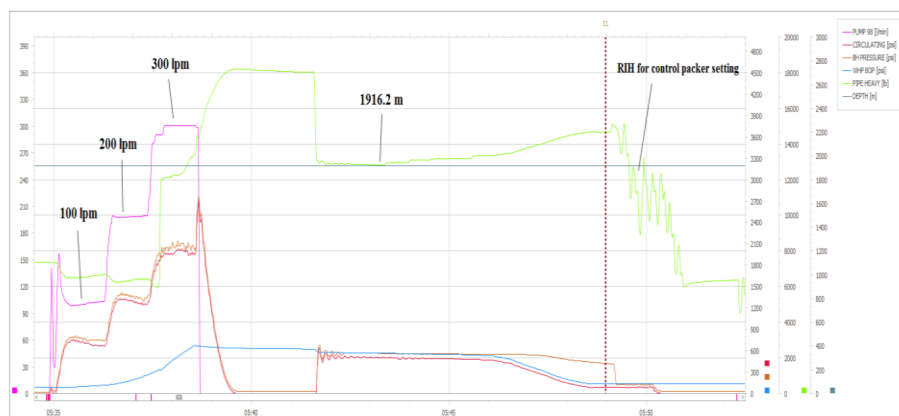
Although not required during the operation, the predefined contingency procedure in case of deflation failure involved the application of a controlled overpull, incrementally increased in 5,000 lbs steps, up to a maximum of 33,460 lbs. This measure was designed to provide mechanical assistance to the retraction of the packer elements, should passive deflation through the bleed orifice and weep hole prove insufficient.

The described procedure was repeated at each subsequent zone, with identical inflation, injection, shut-in, and deflation sequences. Prior to re-engaging inflation at new depths, the CT string was always conditioned with water to prevent circulating viscous fluids through the inflation ports, which could otherwise compromise the packer elements—especially in front of open perforations.

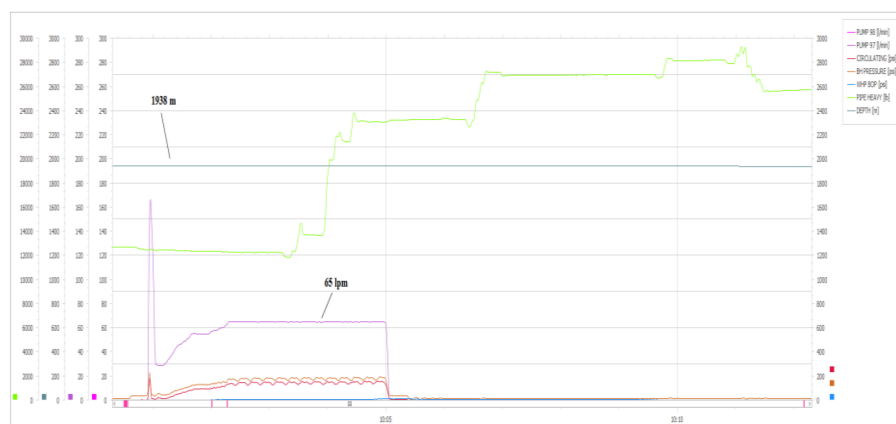
Across all stages, real-time downhole telemetry provided essential feedback for decision-making, confirming packer status, pressure behavior, and tool motion. This significantly improved execution accuracy and reduced risk of non-productive time.



Graph 1—Inflation step.



Graph 2—Mini-frac step.



Graph 3—Deflation step.

Upon conclusion of the final mini-frac, the tool was deflated, and the CT was pulled out of hole to download the memory gauge data and inspect the straddle rubber element conditions.

Evaluation and Conclusion

Upon completion of the operation, the packer system was pulled out of hole and subjected to full post-job inspection, along with the retrieval and download of downhole data from the memory gauges. Both the mechanical inspection and the data analysis confirmed excellent tool performance and data integrity, meeting all operational and quality expectations.

Particular attention during the post job performance evaluation was given to the inflatable rubber elements, which had been the primary concern due to deployment in pre-perforated casing with potential for damage. Visual inspection revealed the elements to be in excellent condition, with only minor surface abrasions observed in localized areas, as shown in the below pictures taken during the post job inspection.



Figure 5—Rubber Status Post Intervention

These superficial marks confirmed effective downhole performance, and more importantly, demonstrated the resilience and robustness of the reinforced rubber design, which maintained full sealing capability throughout all stages of testing operation.

The operation clearly validated the feasibility of performing mini-frac interventions with reinforced inflatable packers rated up to 10,000 psi, even when deployed across pre-perforated casing. Crucial to this success was the strict control of fluid sequencing: the inflatable elements were exposed only to water during both inflation and deflation stages, preventing viscosity-induced retention or damage. Additionally, the custom-formulated viscous fracturing fluid, specifically designed and laboratory-tested to pass through

the small internal clearances of the straddle tool, functioned as intended. It delivered the desired fracture propagation behavior without obstructing tool actuation or compromising system reliability.

This field execution confirmed the viability of high-pressure, rig-less mini-frac operations in challenging downhole environments, leveraging an optimized combination of intelligent CT deployment, reinforced inflatable packer design, and tailored fluid engineering.

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